Support-Two Exactness and Regime Geometry of Shared-Tag TARE Compilation

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Abstract

Shared-Tag Tag-and-Restore (TARE) compilation exposes a coupled design space of auxiliary anticommuting frames, target assignments, shared Tags, and Restore factoring. We study the frozen six-term R6M grammar under its unit support-count objective and separate a durable structural theorem from a deliberately refutable closed-form classification. A complete local audit checks 688,041,472 configurations with zero violations, but its declared Tag-coupling gap is real: exact counterexamples successively reveal Tag-anchor splitting, a central frame-for-Tag borrow, an out-of-support phantom borrow, and a weight-two-Tag plus phantom-borrow hybrid. The invariant result is stronger than any one closed form. A machine-checked all-n exchange theorem proves that frame support at least three never strictly pays, so the unrestricted exact optimum always lies in the support-≤ 2 family D^{++}. A theorem-backed static evaluator of that family has zero error on 9,547 compared instances. By contrast, progressively simpler regime formulas have been honestly refuted: the original three-family identity fails on a fresh $n = 3$ instance, its enlarged-borrow repair is later defeated by 64 hostile hybrid witnesses, and the current all-n finite-basis proof remains one pinned comm-$s2$ lemma short of closure. On recorded H4, equilibrium-N2, and prospectively staged Benzene matchings, the natural weight-one donor family is nevertheless exactly optimal under the frozen objective. The paper therefore contributes an exact support normal form and a refutation-guided map of the remaining low-support coupling geometry, not a claim that a fixed finite trade vocabulary is already complete. All statements are bounded to the frozen grammar/objective; no full-circuit, hardware, or global block-encoding optimality is claimed.

Keywords: quantum compilation; block encoding; TARE; exact optimization; regime geometry; machine-checked theorem.

1 Introduction

Exact compilation is often framed as a search problem: define a sufficiently rich family and optimize inside it. That framing leaves a second scientific question unanswered. Which parts of the search space are actually necessary, which structural restrictions are exact, and what mechanism explains each failure of a restricted family? Shared-Tag TARE is a useful setting for this question because its auxiliary frame, shared Tag, target assignment, and Restore terms interact in a small but nonseparable grammar.

This paper asks one bounded question: for the frozen six-term R6M shared-Tag grammar and unit support-count objective, what support is needed for an exact optimum, and how should the residual low-support regimes be characterized? The distinction between those two clauses is essential. Support exactness is now theorem-grade. Closed-form regime completeness is not.

The research history is itself informative. A weight-one common-anchor family first appeared to capture the exact optimum on the open chemistry subjects. A local dominance audit then explained why extra frame support is expensive, but prospectively declared that Tag repair remained outside the local inequality. Exact search found its first counterexample precisely there. Each subsequent restricted closure was frozen before evaluation and was allowed to fail. Those failures did not weaken the support theorem; they progressively identified the coupling configurations that live inside the theorem's support-two world.

The resulting contribution has three layers. First, a machine-checked all-n theorem proves $C_{\mathrm{DP}} = C_{D^{++}}$: support greater than two is unnecessary for this grammar/objective. Second, exact counterexamples map four currently known coupling configurations—split, borrow, phantom borrow, and split+borrow hybrid—while a current classification proof records one explicit lemma-open sector instead of declaring premature closure. Third, named Hamiltonian batches provide applied grounding: all recorded H4, equilibrium-N2, and prospectively staged Benzene matchings fall in the donor-exact region under this objective.

The paper is intentionally narrower than a global block-encoding claim. It does not assert full-circuit optimality, hardware advantage, objective robustness, or an all-n finite regime vocabulary. Those stronger statements require different evidence and, for reweighted objectives, are already known to be false.

2 Methods and evidentiary contract

2.1 Frozen grammar and objective

An instance consists of six non-identity Pauli targets grouped into three ordered two-target blocks. Each block chooses an anticommuting frame pair, a target permutation, and a central branch. A shared Tag must realize the same two branch labels across all three blocks. Restore strings repair each target from its assigned frame representative. The frozen R6M cost is the sum of block Uanti support cost, twice the shared-Tag support, and the donor-owned three-way Restore-factor cost. The exact serialization and arithmetic are those used by the committed R6M/R6N–R6S analyzers; this manuscript does not redefine them post outcome.

2.2 Referee and restricted families

C_{DP} denotes the unrestricted proof-carrying exact dynamic-programming optimum. The principal restricted families are the original common-anchor weight-one donor family, the split-anchor D^+ family, and D^{++}, which allows frame Paulis of global support at most two. Later closed-form borrow families are explanatory subfamilies of the same frozen grammar. Family containment supplies one-sided inequalities, while equality against the exact referee is established only where a theorem or stated finite comparison authorizes it.

2.3 Prospective refutation discipline

Every family-closure claim used in the discovery ladder was frozen before its outcome. A strict referee gap is therefore a result, not an invalid run. Violating instances are serialized and replayed through independent witness checks. This distinction prevents a counterexample from being hidden by silently enlarging a family after inspection.

2.4 Status vocabulary

We use four evidence labels. An *exact counterexample* proves that its specific closure statement is false. A *complete-domain machine check* exhausts a finite local domain. A *machine-evidenced panel result* applies only to the enumerated instances. An *all-n machine-checked theorem* is promoted only when the finite lemmas are coupled to the stated symbolic/combinatorial composition argument. These labels are not interchangeable.

2.5 Chemistry and prospective subjects

H4 and equilibrium N2 contribute frozen six-term DUCC batches and all 15 perfect matchings per batch. Benzene was selected by a rule frozen before its result, and its predicted costs/regime were digest-stamped before the exact referee ran. High-n D^{++} chemistry values are used only when justified by an exact containment pinch; they are not described as a direct exhaustive sweep.

3 Results

3.1 Local dominance is strong but not jointly complete

The R6N audit checks three local inequalities over 688,041,472 configurations with zero violations. The R6M primary component alone covers 536,870,912 configurations; the letterwise Restore-factor exchange covers 175,616; the rank-2 companion grammar contributes 150,994,944. These checks explain why frame support is expensive, but the protocol explicitly excluded the cost of repairing a shared Tag after changing frame anchors. That declared gap becomes the first discovery surface.

3.2 Two initial exact refutations

The common-anchor weight-one family is refuted on the frozen $n = 2$ panel instance conventionally named `n2_b`: an exact split-anchor realization uses a weight-two shared Tag and costs 8 rather than the donor family's 9. Enlarging to D^+ repairs that witness but is itself refuted. On the minimal structured-$n = 2$ borrow example, a support-two frame on the cheap central branch preserves a weight-one Tag and improves Restore alignment, giving exact cost 5 rather than 6. Across the original R6O panels, D^+ is strictly above the exact referee on 486 of 9,261 structured instances and 73 of 240 seeded random instances.

3.3 Support two is exact for all n

The central structural result is the R6S composition theorem. It proves that support at least three never strictly pays under the frozen R6M unit objective, yielding

$$C_{\mathrm{DP}}(t) = C_{D^{++}}(t) \qquad \text{for every admitted instance } t \text{ and every } n.$$

The proof couples a finite local inequality to a zero-sum/pigeonhole exchange argument. The local inequality is exhaustively checked over 18,432 cases; the associated combinatorial census covers 43,688 class tuples. The exceptional patterns occur exactly at support two, which is why the theorem removes support three and above without erasing the observed support-two borrow configurations.

QG-5b then implements a static evaluator $F_2(t) = C_{D^{++}}(t)$ with no unrestricted-DP call in its forecast path. It has zero error on 9,547 compared instances. Its exactness is theorem-backed; the benchmark is a consistency check of the implementation rather than the source of the theorem.

3.4 Closed-form regime formulas remain refutable

The earlier formula $\min(C_{R6L}, C_{D^+}, f_B)$ matched its original 9,771-classification corpus but fails on a later fresh $n = 3$ instance with exact cost 10 versus forecast 11. Enlarging the borrow home domain to B' repairs that witness and closes all QG-5b verified rows. QG-7 then freezes an adversarial search specifically against B' and finds 64 exact fourth-configuration witnesses among 740 instances. Their characteristic form combines a multi-anchor weight-two Tag with a phantom support-two borrow; a representative has $C_{\mathrm{DP}} = C_{D^{++}} = 7 < 8 = \min(C_{D^+}, f_{B'})$.

The subsequently frozen hybrid family B'' covers all 64 witnesses and closes 10,481 QG-7b compared instances, including a new anti-B'' panel, with no fifth configuration found on those panels. This is not promoted to an all-n finite-basis theorem. QG-7c closes the weight-≥ 3 Tag rung and three of four remaining shape classes, but leaves the pinned `comm-s2` sector lemma-open. Hostile realized examples of that sector show no new cost gap, so the current uncertainty is a proof obligation rather than a demonstrated fifth regime.

3.5 Chemistry lies in the donor-exact region

On all recorded H4 and equilibrium-N2 matchings, the exact optimum equals the natural weight-one donor family under the unit objective. The prospective Benzene study adds 15 matchings whose donor-exact regime and exact costs were staged before exact computation and then confirmed 15/15. These observations do not prove that chemistry generally lies in the donor region; they establish the statement only for the named frozen batches.

3.6 Applied grounding

The R4B split-TARE coefficient result reports zero hostile failures over 8,700 partition evaluations and isolates a coefficient-coordinate theorem. The R4D blob-locked H2O run reduces its frozen structural TARE coordinate from 8,078 to 4,972 at the stated normalization overhead. Both results motivate the relevance of structural-cost optimization while explicitly stopping short of full-circuit or hardware optimality.

4 Discussion

The main lesson is that support complexity and regime-description complexity are different objects. The support theorem is stable under every later refutation described here: QG-5 and QG-7 discover new configurations *inside* D^{++}, not counterexamples to D^{++}. A compact classifier can therefore fail while the exact support normal form remains intact.

This distinction changes how the chemistry result should be read. The natural donor family is exactly optimal on the named batches, but the reason is not that unrestricted optimization is vacuous. Nearby synthetic and hostile instances exhibit exact improvements through several low-support coupling patterns. Donor exactness is a property of a regime, not a universal property of the grammar.

The refutation ladder also provides a methodological benefit. Each failed closed form localizes what the previous language could not express: first split anchors, then cheap central support, then out-of-support homes, then hybrid multi-anchor Tag/phantom interactions. The current QG-7c boundary makes the remaining theorem debt explicit. A future proof may close the pinned `comm-s2` sector without adding a fifth configuration, or an exact witness may force another enlargement. Either outcome is compatible with the support-two theorem.

Finally, objective dependence must remain visible. QG-2 already exhibits a reweighted objective for which support three can pay. The theorem in this paper is therefore about a specific compilation family *and* objective, not an intrinsic statement that TARE compilation is always support-two.

5 Related work and donor boundary

The closest literature separates into three axes. First, block-encoding and qubitization/QSVT provide the broader algorithmic setting in which linear combinations and signal transformations are represented, but they do not specify the frozen shared-Tag TARE grammar studied here [1, 2]. Second, the TARE construction descends from a stabilizer-formalism block-encoding method that transforms Pauli strings into an anticommuting family and applies a correction/Restore transformation [3]; anticommuting-unitary partitioning is also established in quantum measurement work [4]. Third, Pauli-level compiler representations and optimization are established, including PCOAST [5] and recent global binary-symplectic simplification for Hamiltonian simulation [6]. These are donors, foundations, or adjacent compiler systems rather than evidence for the exact support theorem below.

Accordingly, the TARE primitive, Tag/Restore identity, freely chosen anticommuting auxiliary families, unitary-partitioning machinery, block-encoding/QSVT foundations, Clifford/symplectic synthesis, and global Pauli compilation receive no novelty credit in this manuscript. The candidate residual is only the exact structure earned by the committed evidence: for one frozen six-term shared-Tag grammar and one unit support-count objective, the all-n support-two normal form is exact and sharp, while exact counterexamples delimit narrower regime families and leave one explicit finite-basis lemma open.

A submission-date refresh on 2026-08-31 checked the donor block-encoding line, anticommuting-unitary partitioning, Pauli compiler frameworks, and current symplectic compilation work together with foundational block-encoding literature. No retrieved source in that bounded search states the same R6M support-two theorem. This is not an exhaustive priority certificate. A closer donor found during review must narrow the residual rather than be rhetorically excluded, and the paper therefore avoids 'first', 'unique', or externally certified novelty language.

The negative result history is part of the positioning rather than an embarrassment to be removed. The original three-family identity fails; an enlarged-borrow repair is later defeated by 64 hostile hybrid witnesses; and the current all-n finite-basis classification remains one pinned comm-$s2$ lemma short of closure. Those failures do not weaken the support-two theorem, but they prohibit a complete fixed-vocabulary claim. Likewise, finite audits and application matchings corroborate the scoped structure without becoming full-circuit, hardware, runtime, or global block-encoding claims.

6 Limitations

First, the all-n theorem is objective-scoped. Reweighting central, non-central, Restore, or Tag costs can change the normal form; QG-2 contains an explicit support-three counterexample under one such objective. Second, the grammar is frozen to the studied shared-Tag six-term construction. Larger Tag ranks, other block grammars, other term counts, and different factor rules require separate proofs.

Third, the finite closed-form regime vocabulary is not yet an all-n theorem. B'' closes the stated 10,481-instance corpus, and QG-7c reduces the proof to one pinned sector, but absence of a fifth configuration on hostile finite panels is not proof of absence. Fourth, the chemistry evidence covers named H4, equilibrium-N2, and Benzene batches only. It cannot support a prevalence claim over chemistry Hamiltonians.

Fifth, the cost is a structural compilation coordinate rather than a full hardware resource model. State preparation, architecture, routing, fault-tolerant synthesis, measurement, and system-level scheduling can change the practical optimum. The R4B/R4D results narrow this gap but do not remove it.

Finally, this work belongs to a single-author research programme using AI assistance, so the verification described here is author-side throughout and lacks the independent perspective of multi-author or externally replicated work.

7 Conclusion

For the frozen R6M shared-Tag TARE grammar under its unit support-count objective, support two is enough for exact optimization at every qubit count: $C_{\mathrm{DP}} = C_{D^{++}}$. That theorem survives a sequence of deliberately hostile closed-form failures. The failures reveal an increasingly precise low-support regime geometry, from split anchors and central borrows to phantom and hybrid configurations, while the current all-n finite-basis proof remains one named lemma short of closure.

The durable conclusion is therefore not that TARE has exactly two, three, or four trades. It is that unrestricted optimization collapses to an exact support-two normal form, and that the remaining task is classification within that normal form. On the named chemistry batches, the natural donor family lies exactly in the optimum regime; beyond them and beyond the frozen objective, no broader optimality claim is made.

8 Reproducibility

Every quantitative statement is bound to committed protocols, result JSON, and generating code. The primary chain is R6N–R6S under `research/extensions/orion-q/` and `development/orion-q-max-r0/`; the current classification stress tests are QG-5b and QG-7/QG-7b/QG-7c under `research/extensions/orion-qg/` and `development/orion-qg-regime-geometry/`. The applied grounding uses the R4B and R4D receipts.

Reproduction should verify receipt digests before recomputation, run the committed analyzers without changing frozen families or gates, and compare canonical outputs byte-for-byte where the receipt contract declares deterministic replay. A counterexample must remain serialized even after a later enlarged family covers it. High-n chemistry equalities derived by an exact containment pinch must be reported as such.

A concise verification order and integrity rules are given in the paper-level `REPRODUCE.md`. The protected stretched-N2 subject is not required for any claim in this manuscript.

9 Ethics, safety and resource statement

This work uses no human participants, animals, clinical data, or personal data. The principal research-integrity risks are computational rather than biomedical: post-outcome family expansion, erasure of negative results, accidental access to protected subjects, and promotion of a receipt beyond its authority. The protocol/receipt design addresses these risks by freezing comparison families and gates before outcome access, retaining exact refutations, and keeping the protected stretched-N2 subject sealed.

The largest complete local audit enumerates 688,041,472 configurations. Later exact and hostile searches are finite and deterministic under their stated caps/seeds. Computational expense is reported as a property of the verification procedure, not as evidence for scientific importance. No receipt in this paper grants physical quantum advantage, novelty, adoption, or deployment authority.

Author contribution and AI assistance. Sze Chun Yiu is responsible for the research design, implementation, evidence interpretation, claim decisions, and final manuscript. Large-language-model assistance was used during publication closure for editorial refinement, bibliographic research, consistency checking, and package preparation. It did not receive independent scientific authority: every retained claim, citation, and boundary was reviewed against the committed evidence, and the author accepts responsibility for the final text and computations.

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